

Power Plant Engineering
Laboratory Manual
Course Code: ME407.01
B.Tech 8th Sem

ME407.01 Power Plant Engineering	
CO1	Describe sources of energy and types of power plants.
CO2	Analyze the performance of diesel powered thermal power plant.
CO3	Describe basic working principles of gas turbine.
CO4	List the principal components and types of nuclear reactors.
CO5	List types, principles of operations, components and applications of steam turbines, steam generators, condensers, feed water and circulating water systems.
CO6	Estimate different efficiencies associated with power plant systems.
CO7	Analyze economics of power generation.

List of Experiments (ME 407.01 PPE)		
Sr. No.	Title	Course Outcomes
1	To study of modern steam power plant.	CO1
2	To Study about the Various Types of Fuel & Ash Handling Systems.	CO1, CO5
3	To study about different types of dust collectors and pulverized fuel burners.	CO1, CO5
4	To study about nuclear power plant.	CO4
5	To study of different types of steam turbines.	CO5
6	To study about different types of condensers and cooling towers.	CO5
7	To study about economics of power generation systems.	CO7
8	To study of gas power plant.	CO3, CO6
9	To study of combined steam & gas turbine power plant.	CO1
10	Testing of diesel fired water tube boiler based steam power plant.	CO2, CO6

CERTIFICATE

This is to certify that Mr. /Ms. _____ of
_____ Class, Roll No. _____ Exam No.
_____ has satisfactorily completed his / her term work in
_____ for the term ending
_____ in 20__ / 20__.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY, CHANGA – 388 421

Date :

Sign of the Faculty Head of the Department

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EXPERIMENT NO. 1 STUDY OF STEAM POWER PLANT.

AIM:

To study of modern steam power plant.

OBJECTIVES:

1. Study about the layout of steam power plant.
2. Study about the rankine cycle and different components steam power plant.

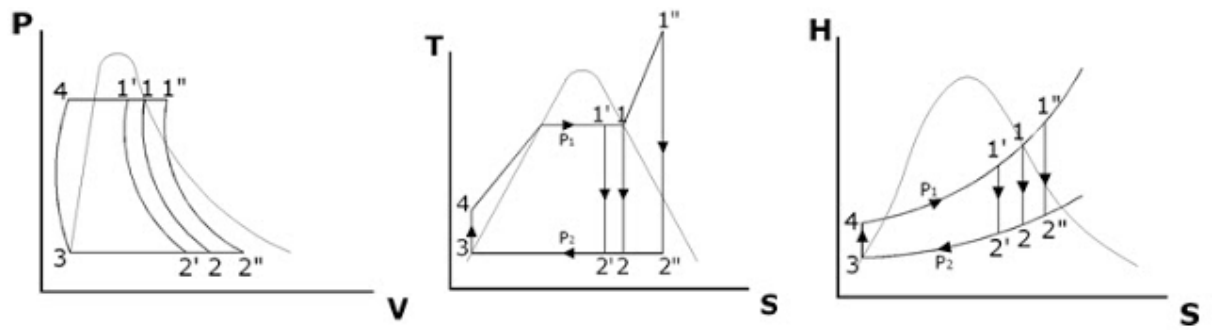
THEORY:

1. Modern Steam Power Plant and its Operating Cycle.

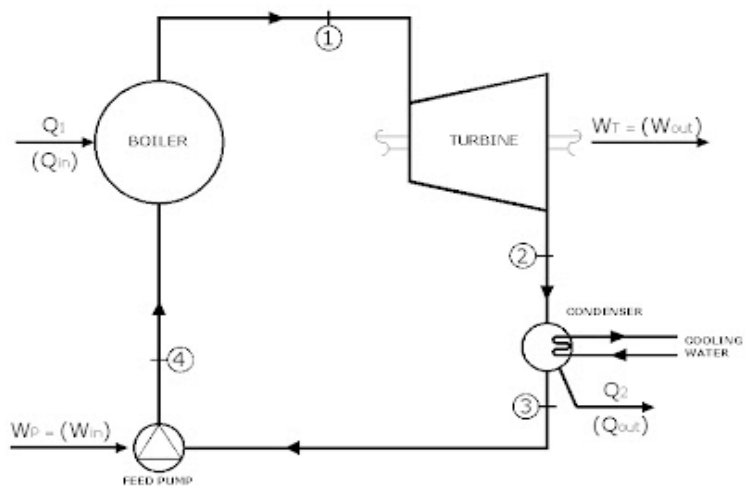
The general layout of the modern power plant consists of mainly four circuits which are

- Coal and ash circuit.
- Air and gas circuit.
- Feed water and steam flow circuit.
- Cooling water circuit.

A thermal power station using steam as working fluid works basically on the Rankine cycle. Steam is generated in a boiler, expanded in a prime mover and condensed in a condenser and fed into the boiler again with the help of pump. However, in practice, there are numerous modifications and improvements in this cycle with the aim of affecting heat economy and to increase the thermal efficiency of the plant.



P-V, T-S and H-S diagram of simple Rankine cycle



Simple Rankine cycle

EXPERIMENT NO. 2

STUDY OF DIFFERENT TYPES OF FUEL AND ASH HANDLING SYSTEMS

AIM:

To Study about the Various Types of Fuel & Ash Handling Systems.

OBJECTIVES:

3. Study about different coal and ash handling systems.

THEORY:

Coal is the most important fuel factor for running the steam power plant, so it needs to have a care of the fuel from its initial stage to the final stage of using it as a fuel into the Boiler. And so for that it needs proper organized handling system from the transportation to its final destination in the powder form.

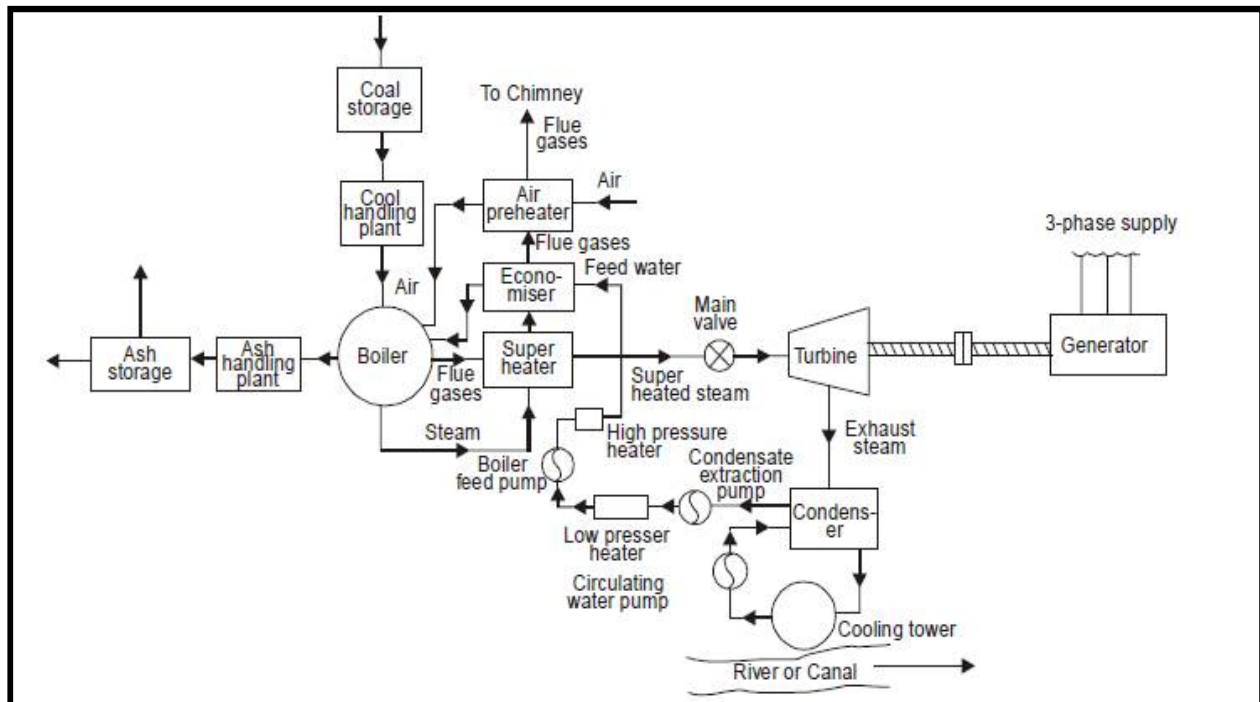


Fig 1 Layout of Power plant

There are mainly four units which must be installed in any power plant are:

1. Feed water and steam flow circuit
2. Coal and ash circuit
3. Air and gas circuit
4. Cooling water circuit.

In above system Coal & ash circuit is most important role into the power plant. On basis of these coal & ash handling system is generated for the proper operation.

COAL HANDLING:

Coal delivery equipment is one of the major components of plant cost. The various steps involved in coal handling are as follows:

1. Coal delivery
2. Unloading
3. Preparation
4. Transfer
5. Outdoor storage (Dead Storage)
6. Covered storage (Live Storage)
7. In plant handling
8. Weighing and measuring
9. Feeding the coal into furnace.

DEWATERING OF COAL

Excessive surface moisture of coal reduces and heating value of coal and creates handling problems. The coal should therefore be dewatered to produce clean coal. Cleaning of coal has the following advantages:

1. Improved heating value.

2. Easier crushing and pulverizing
3. Improved boiler performance
4. Less ash to handle.
5. Easier handling.
6. Reduced transportation cost.

PULVERIZED COAL:

Coal is pulverized (powdered) to increase its surface exposure thus permitting rapid combustion. Efficient use of coal depends greatly on the combustion process employed.

For large scale generation of energy the efficient method of burning coal is confined still to pulverized coal combustion. The pulverized coal is obtained by grinding the raw coal in pulverizing mills. The various pulverising mills used are as follows:

- | | |
|-----------------------|----------------|
| 1. Ball mill | 2. Hammer mill |
| 3. Ball and race mill | 4. Bowl mill. |

The essential functions of pulverising mills are as follows:

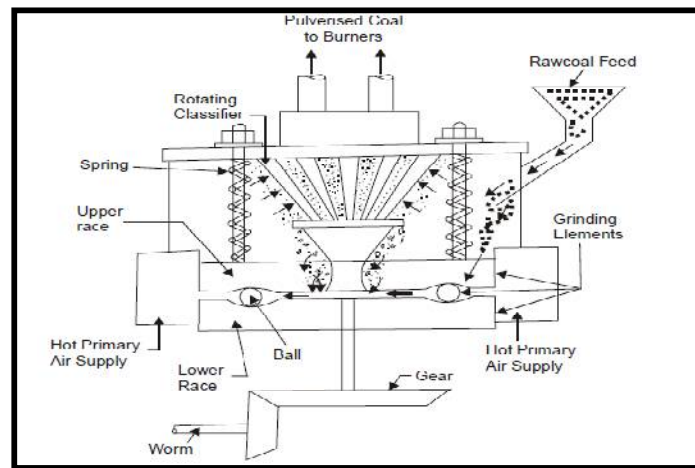
1. Drying of the coal
2. Grinding
3. Separation of particles of the desired size.

The coal pulverising mills reduce coal to powder form by three actions as follows:

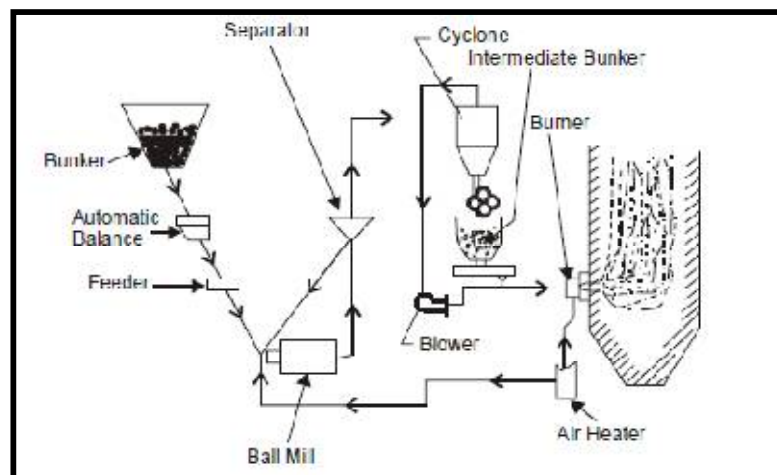
1. Impact
2. Attrition (abrasion)
3. Crushing.

Most of the mills use all the above mentioned all the three actions in varying degrees. In impact type mills hammers break the coal into smaller pieces whereas in attrition type the coal pieces which rub against each other or metal surfaces to disintegrate. In crushing type mills coal caught between metal rolling surfaces gets broken into pieces. The crushing mills use steel balls in a container. These balls act as crushing elements.

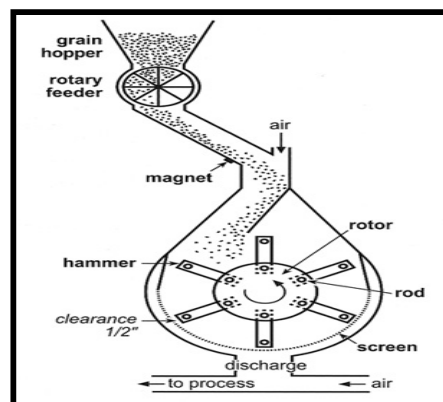
BALL MILL:



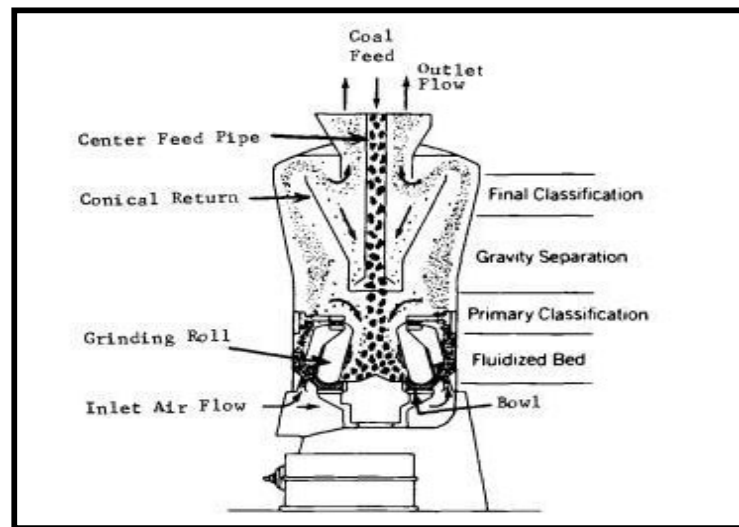
BALL & RACE MILL:



HAMMER MILL:



BOWL MILL:



ASH DISPOSAL:-

A large quantity of ash is, produced in steam power plants using coal. Ash produced in about 10 to 20% of the total coal burnt in the furnace. Handling of ash is a problem because ash coming out of the furnace is too hot, it is dusty and irritating to handle and is accompanied by some poisonous gases.

It is desirable to quench the ash before handling due to following reasons:

1. Quenching reduces the temperature of ash.
2. It reduces the corrosive action of ash.
3. Ash forms clinkers by fusing in large lumps and by quenching clinkers will disintegrate.
4. Quenching reduces the dust accompanying the ash.

Handling of ash includes its removal from the furnace, loading on the conveyors and delivered to the fill from where it can be disposed off.

ASH HANDLING EQUIPMENT

Mechanical means are required for the disposal of ash. The handling equipment should perform the following functions:

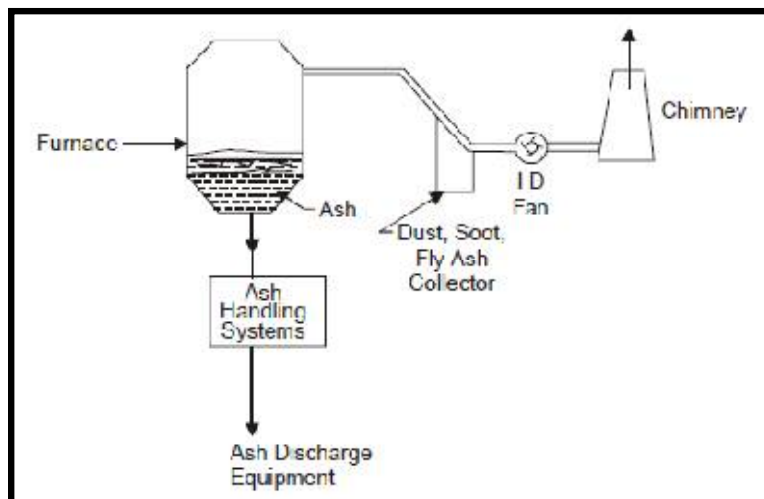
- (1) Capital investment, operating and maintenance charges of the equipment should be low.
- (2) It should be able to handle large quantities of ash.
- (3) Clinkers, soot, dust etc. create troubles, the equipment should be able to handle them smoothly.
- (4) The equipment used should remove the ash from the furnace, load it to the conveying system to deliver the ash to a dumping site or storage and finally it should have means to dispose of the stored ash.
- (5) The equipment should be corrosion and wear resistant.

Fig. shows a general layout of ash handling and dust collection system. The commonly used ash handling systems are as follows:

1. Hydraulic system
2. Pneumatic system
3. Mechanical system.

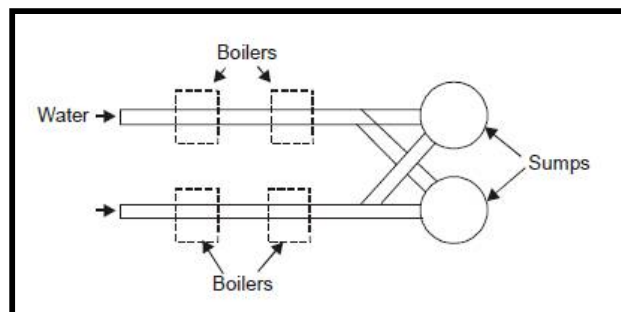
The commonly used ash discharge equipment is as follows:

- a. Rail road cars
- b. Motor truck & Barge



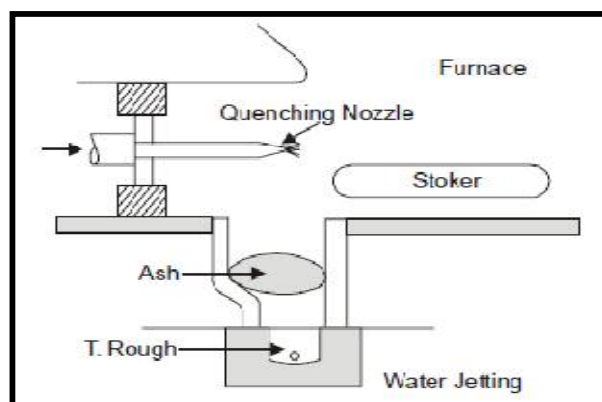
The various methods used for the disposal of ash are as follows :

1. **Hydraulic System**. In this system, ash from the furnace grate falls into a system of water possessing high velocity and is carried to the sumps. It is generally used in large power plants. Hydraulic system is of two types namely low pressure hydraulic system used for continuous removal of ash and high pressure system which is used for intermittent ash disposal. Fig. shows hydraulic system.

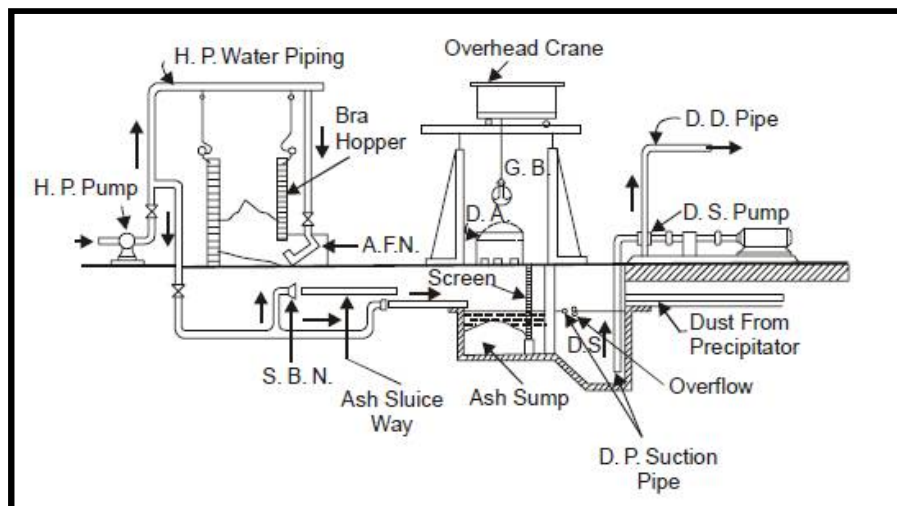


In this method water at sufficient pressure is used to take away the ash to sump. Where water and ash are separated. The ash is then transferred to the dump site in wagons, rail cars or trucks. The loading of ash may be through a belt conveyor, grab buckets. If there is an ash basement with ash hopper the ash can fall, directly in ash car or conveying system.

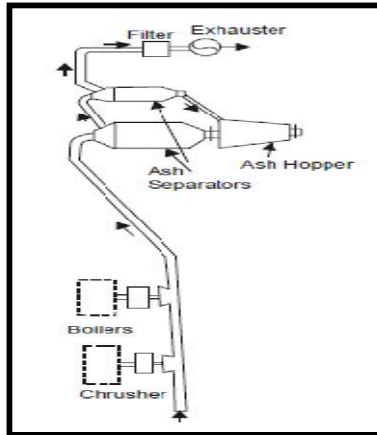
2. **Water Jetting**. Water jetting of ash is shown in Fig. In this method a low pressure jet of water coming out of the quenching nozzle is used to cool the ash. The ash falls into a trough and is then removed.



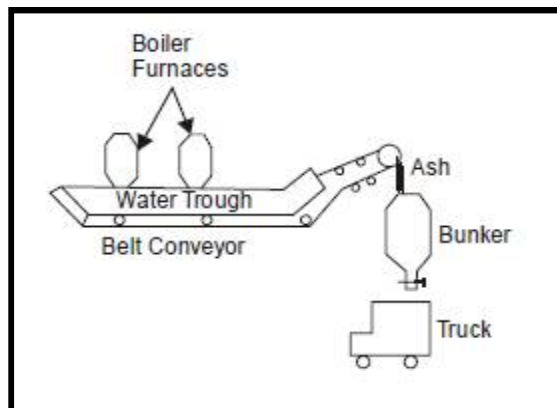
3. **Ash Sluice Ways and Ash Sump System.** This system shown diagrammatically in Fig. used high pressure (H.P.) pump to supply high pressure (H.P.) water-jets which carry ash from the furnace bottom through ash sluices (channels) constructed in basement floor to ash sump fitted with screen. The screen divides the ash sump into compartments for coarse and fine ash. The fine ash passes through the screen and moves into the dust sump (D.S.). Dust slurry pump (D.S. pump) carries the dust through dust pump (D.P), suction pipe and dust delivery (D.D.) pipe to the disposal site. Overhead crane having grab bucket is used to remove coarse ash. A.F.N represents ash feeding nozzle and S.B.N. represents sub way booster nozzle and D.A. means draining apron.



4. **Pneumatic system.** In this system ash from the boiler furnace outlet falls into a crusher where larger ash particles are crushed to small sizes. The ash is then carried by a high velocity air or steam to the point of delivery. Air leaving the ash separator is passed through filter to remove dust etc. so that the exhaustor handles clean air which will protect the blades of the exhaustor.



5. **Mechanical ash handling system.** Fig. shows a mechanical ash handling system. In this system ash cooled by water seal falls on the belt conveyor and is carried out continuously to the bunker. The ash is then removed to the dumping site from the ash bunker with the help of trucks.



Questions:

1. Name the various methods of ash handling. Describe the pneumatic system of ash handling. Why it is essential to quench the ash before handling?
2. Describe the various methods used to fire pulverized coal. Make a neat sketch of ball and Race mill and explain its working
3. Name the different types of coal-pulverising mills. Describe Ball-Mill.

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EXPRIMENT NO.3

STUDY ABOUT DIFFERENT TYPES OF DUST COLLECTORS AND PULVERISED FUEL BURNERS

AIM:

To study about different types of dust collectors and pulverized fuel burners.

OBJECTIVE:

To study about different dust collectors and fuel burners.

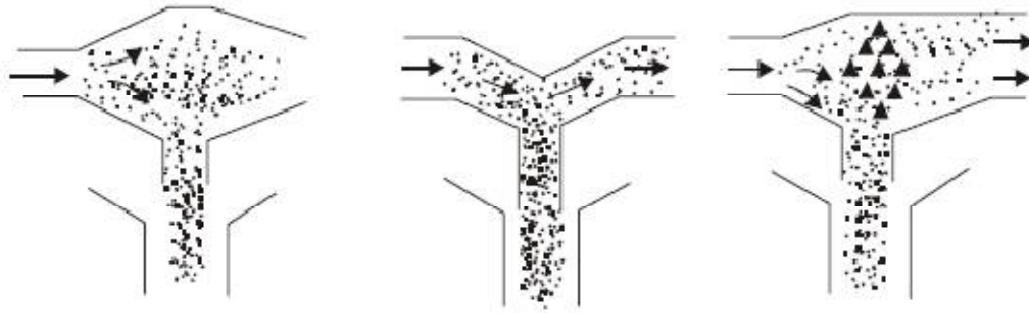
THEORY:

The various types of dust collectors are as follows:

1. Mechanical dust collectors.
2. Electrical dust collectors.

Mechanical dust collectors:

Mechanical dust collectors are sub-divided into wet and dry types. In wet type collectors also known as scrubbers water sprays are used to wash dust from the air. The basic principles of mechanical dust collectors are shown in Fig. As shown in Fig. by increasing the cross-sectional area of duct through which dust laden gases are passing, the velocity of gases is reduced and causes heavier dust particles to fall down. Changing the direction of flow of flue gases causes the heavier particles of settle out. Sometime baffles are provided as shown in Fig. to separate the heavier particles. Mechanical dust collectors may be wet type or dry type. Wet type dust collectors called scrubbers make use of water sprays to wash the dust from flue gases.



Dry type dust collectors include gravitational, cyclone, louvered and baffle dust collectors. A cyclone dust collector uses a downward flowing vortex for dust laden gases along the inner walls. The clean gas leaves from an inner upward flowing vortex. The dust particles fall to the bottom due to centrifuging action.

Cyclone separator:

In this type of mechanical collector, a high velocity gas stream carrying the dust particles enters at high velocity and tangential to the conical shell. This produces the whirling motion of the gas within the chamber and throws heavier dust particles to the sides and fall out of gas stream and is collected at the bottom of the collector. The gas from the conical shell is passed through the secondary chamber.

Electrostatic Precipitators:

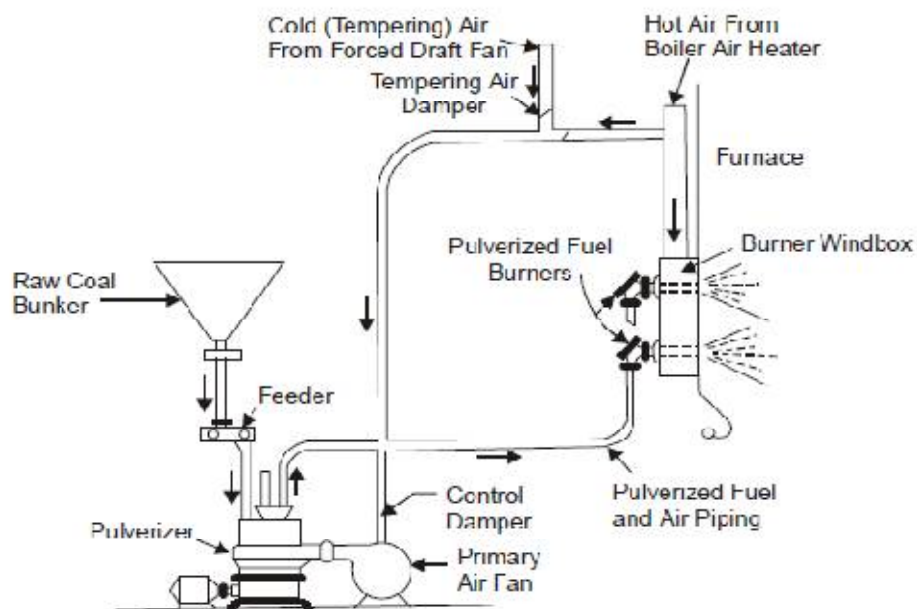
It has two sets of electrodes, insulated from each other that maintain an electrostatic field between them at high voltage. The flue gases are made to pass between these two sets of electrodes. The electric field ionises the dust particle; that pass through it attracting them to the electrode of opposite charge. The other electrode is maintained at a negative potential of 30,000 to 60,000 volts. The dust particles are removed from the collecting electrode by rapping the electrode periodically. The electrostatic precipitator is costly but has low maintenance cost and is frequently employed with pulverised coal fired power stations for its effectiveness on very fine ash particles and is superior to that of any other type.

Pulverised fuel burners:

Burners are used to burn the pulverised coal. The main difference between the various burners lies in the rapidity of air-coal mixing *i.e.*, turbulence. For bituminous coals the turbulent type of burner is used whereas for low volatile coals the burners with long flame should be used.

A pulverised coal burner should satisfy the following requirements:

- (i) It should mix the coal and primary air thoroughly and should bring this mixture before it enters the furnace in contact with additional air known as secondary air to create sufficient turbulence.
- (ii) It should deliver air to the furnace in right proportions and should maintain stable ignition of coal air mixture and control flame shape and travel in the furnace. The flame shape is controlled by the secondary air vanes and other control adjustments incorporated into the burner. Secondary air if supplied in too much quantity may cool the mixture and prevent its heating to ignition temperature.
- (iii) Coal air mixture should move away from the burner at a rate equal to flame front travel in order to avoid flash back into the burner.

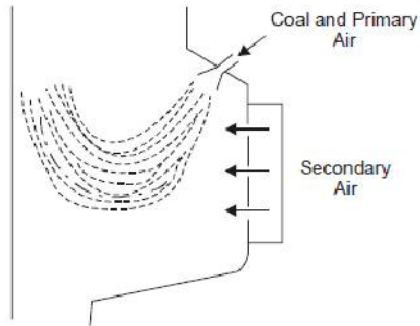


Pulverised Fuel Burner

The various types of pulverized fuel burners are as follows :

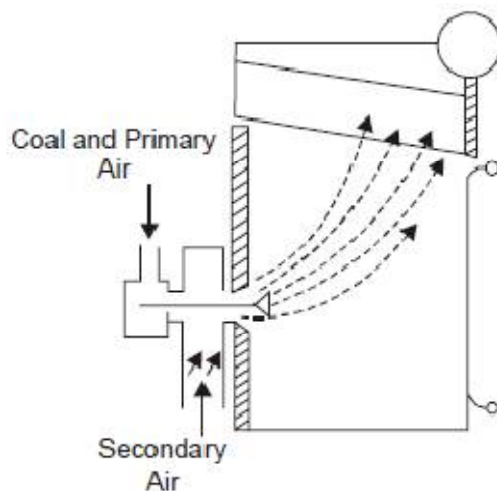
1. Long Flame Burner (U-Flame Burner):

In this burner air and coal mixture travels a considerable distance thus providing sufficient time for complete combustion



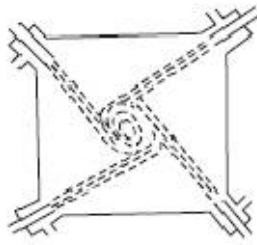
2. Short Flame Burner (Turbulent Burner):

The burner is fitted in the furnace wall and the flame enters the furnace horizontally.



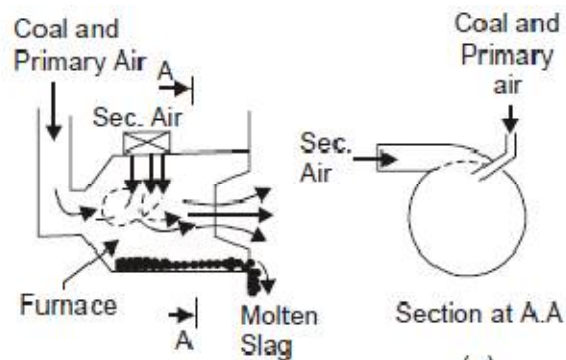
3. Tangential Burner:

In this system one burner is fitted attach corner of the furnace. The inclination of the burner is so made that the flame produced are tangential to an imaginary circle at the centre.



4. Cyclone Burner:

This burner uses crushed coal instead of pulverized coal. Its advantages are as follows:



- (i) It saves the cost of pulverization because of a crusher needs less power than a pulveriser.
- (ii) Problem of fly ash is reduced. Ash produced is in the molten form and due to inclination of furnace it flows to an appropriate disposal system. .

QUESTIONS:

1. Explain Electrostatic precipitator with its sketch.
2. Explain with neat sketch different types of pulverized fuel burners.

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EXPRIMENT NO.4

STUDY ABOUT NUCLEAR POWER PLANT

AIM:

To study about nuclear power plant.

OBJECTIVE:

To study about components of nuclear power plant and its types.

THEORY:

Cheap and abundant power is essential to the modern world in coming years. The nuclear power is not only available in abundance but it is cheaper than the power generated by conventional sources.

One of the outstanding facts about nuclear power is the large amount of energy that can be released from a small mass of active material. Complete fission of one kg of uranium contains the energy equivalent to 3100 tons of coal or 1700 tons of oil.

A nuclear power plant is a place where people make electricity using heat from nuclear reactions. A nuclear power plant has a place where the nuclear reaction happens called a reactor. The plant also has machines which remove heat from the reactor and make electricity. Electricity made by nuclear power plants is called nuclear power. Nuclear power plants are usually located near water to remove the heat the reactor makes. Some nuclear power plants use cooling towers to do this.

Nuclear power plants are powered by Uranium. In a process known as nuclear fission, uranium atoms are split to produce large amount of energy which is eventually converted to

heat. The enormous amount of heat created, boils the water to produce steam, which is used to rotate turbines. These turbines in-turn spin the shaft of the generator. As the generator gets into action, the coils of wire within the generator are spun in a magnetic field to produce electricity.

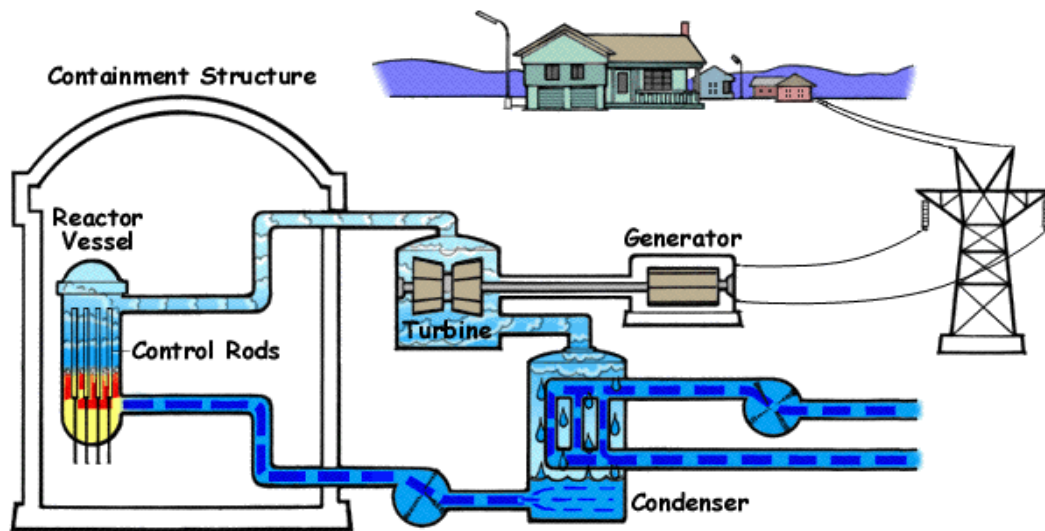
A nuclear reactor maintains and controls the nuclear reaction within the plant to produce energy. In the United States, pressurized water reactors and boiling water reactors are used in nuclear power plants.

Though there are a few security concerns about operations of a nuclear power plants, they are necessary as an alternative energy source, to cope up with the ever increasing energy requirements. Owing to low levels of emission, this cost-effective source of power is steadily becoming a popular source.

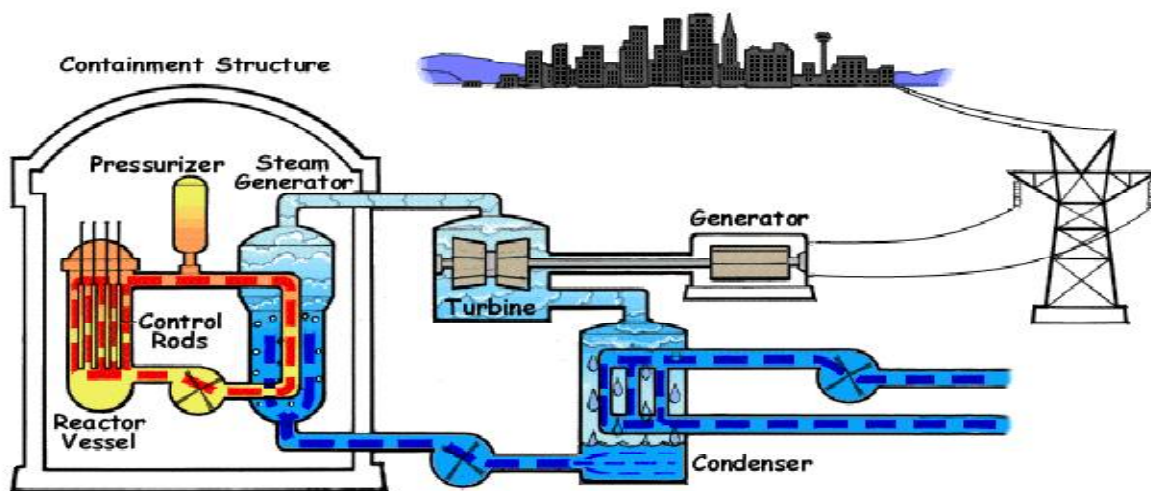
It is estimated that the demand for nuclear energy for production of electricity will increase by 20%, by 2030. Knowing how does a nuclear power plant work, and the safety measures ensured in the process, will perhaps make many people change their critical stance on nuclear plants.

Different types of Nuclear Power Plants:-

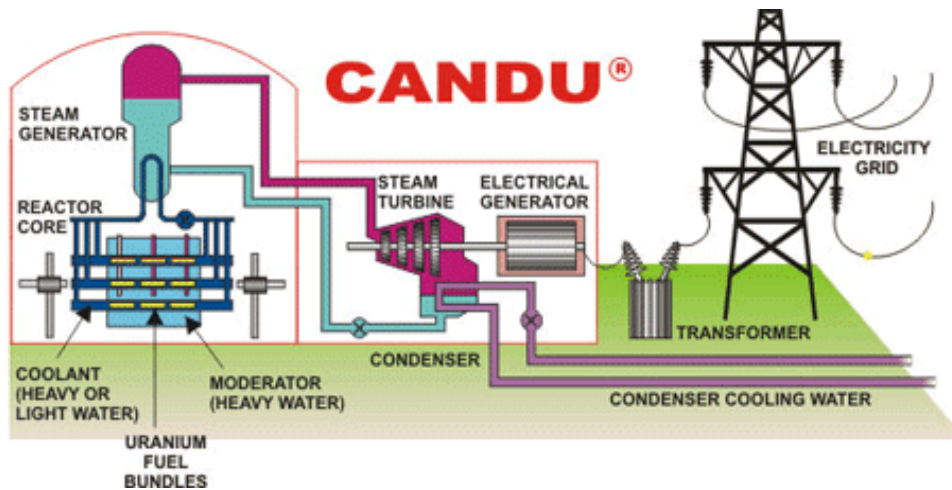
A. Boiling water reactor power plant



B. Pressurized water reactor power plant (PWR)



C. Heavy water reactor power plant (CANDU)



QUESTIONS:

1. Classify different types of nuclear reactors.
2. Explain PWR, BWR and CANDU power plant with its construction and working.

REFERENCES:

1. Power Plant Engineering by P.K.Nag, TMH Publications.
2. Steam and Gas turbines and Power Plant Engineering by R.Yadav, Central publishing house, Allahabad.

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EXPERIMENT NO. 5

STUDY OF STEAM TURBINES AND ITS TESTING

AIM:

To study of different types of steam turbines.

OBJECTIVES:

1. To study about the different types of steam turbines and their applications.
2. To go for testing of given steam turbine for the given power plant and to find out the output for the same.

PRINCIPLE OF OPERATION:-

Steam turbine converts the heat energy of steam into mechanical work. The principle of operation of any turbine depends on Newton's second law of motion. The motive power in a turbine is obtained by the change in momentum of a high velocity jet impinging on a curved blade.

The steam from the boiler is expanded in a nozzle where due to fall in pressure of steam, thermal energy of steam is converted into kinetic energy of steam, resulting in the emission of a high velocity jet of steam which impinges on the moving vanes or blades, mounted on the shaft. It undergoes a change in the direction of motion which gives rise to a change in momentum and force.

CLASSIFICATION OF STEAM TURBINES:-

Steam turbines are classified as

i. On the basis of principle of operation

a) Impulse turbine:-

If the flow of steam through the nozzles and moving blades of a turbine takes place in such a manner that the steam is expanded only in nozzles and the pressure at the outlet side of blade is equal to that at inlet i.e. drop in pressure of steam takes place only in nozzles and not in moving blades; such a turbine is known as impulse turbine.

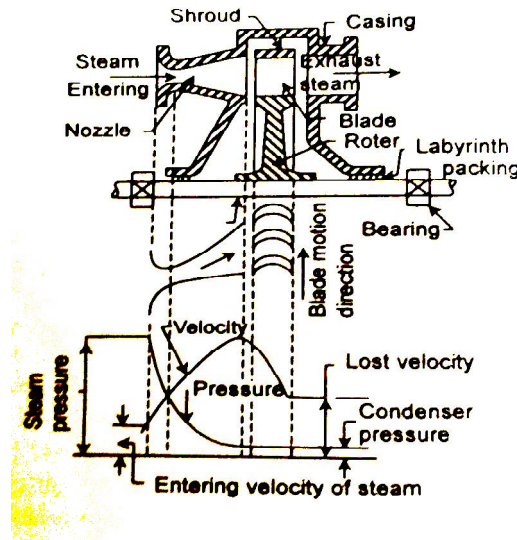
b) Impulse reaction turbine:-

Expansion of steam takes place in nozzles as well as in moving blades. The pressure of steam at the outlet from these moving blades of a turbine is less than that at the inlet side of blades; which causes rise in generation of kinetic energy within the blades, giving rise to reactions and add to the propelling force. Such a turbine is known as impulse reaction turbine.

SIMPLE IMPULSE TURBINE:-

It works on the principle of impulse. It consists of a nozzle or a set of nozzles, a rotor mounted on a shaft, one set of moving blades attached to the rotor and casing.

This type of turbine is generally employed where relatively small power is needed and where the rotor diameter is fairly small. The small rotor gives a very high rotational speed, reaching 30000 rpm.



COMPOUNDING OF IMPULSE TURBINE:-

This method is employed for reducing the rotational speed of the impulse turbine to practical limits. If the high velocity of steam is allowed to flow through one row of moving blades, it produces a rotor speed of about 30000 rpm which is too high for practical use and even leaving loss is too high.

It is therefore essential to incorporate improvements in the simple impulse turbine as to make it more efficient. This is achieved by making use of more than one set of nozzles, blades, rotors in series keyed to a common shaft, so that either the steam pressure or the jet velocity is absorbed by the turbine in stages. This also reduces the leaving loss. This process is called compounding of steam turbine.

1. Compounding of impulse turbines

There are three main types of impulse compound turbines

a) Pressure-compounded impulse turbine

In this type of turbine, compounding is done for pressure of steam only i.e. to reduce the high rotational speed of turbine the whole expansion of steam is arranged in a number of steps by employing a number of simple turbines in series keyed on the same shaft.

Each of these simple impulse turbine consisting of one set of nozzles and one row of moving blades is known as stages of turbine and this turbine consists of several stages.

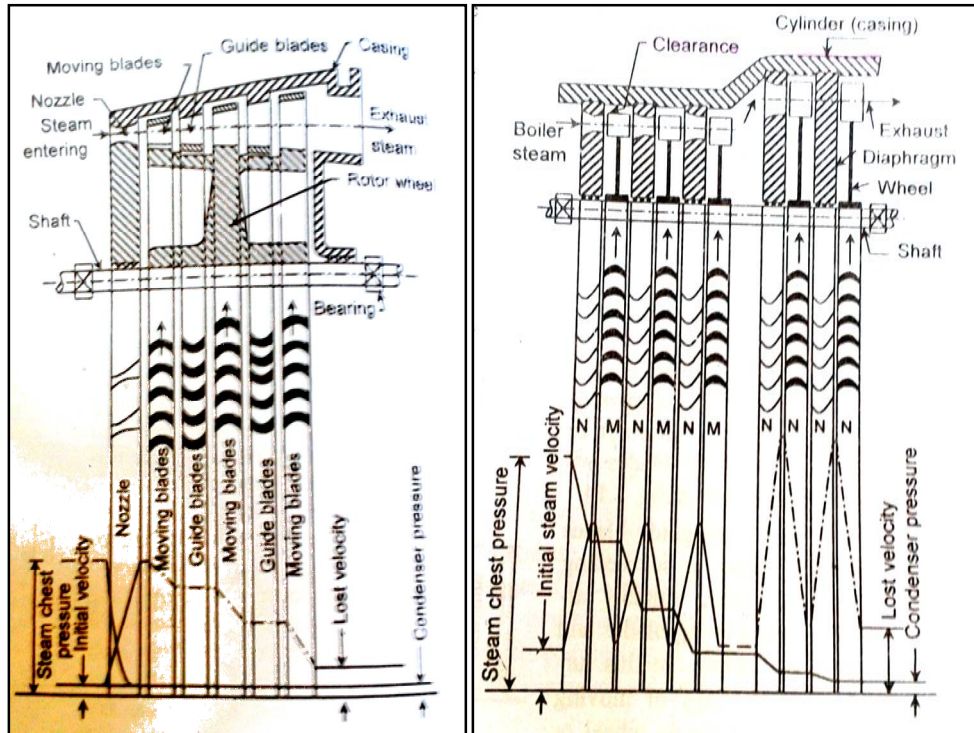
b) Velocity-compounded impulse turbine

In this type of turbine, compounding is done for velocity of steam only i.e. drop in velocity is arranged in many small drops through many moving rows of blades instead of a single row of moving blades.

It consists of a nozzle or a set of nozzles and rows of moving blades attached to the rotor or wheel and rows of fixed blades attached to casing.

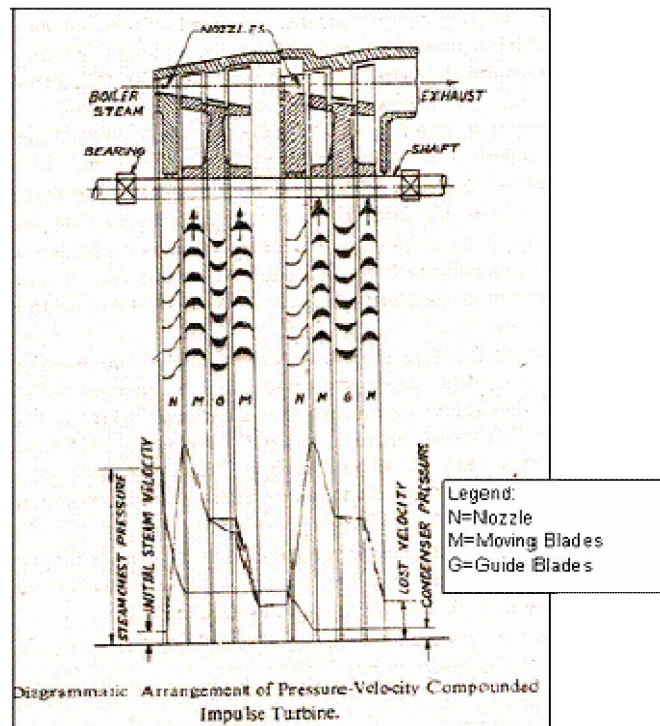
c) Pressure and velocity compounded impulse turbine.

This type of turbine is combination of pressure and velocity compounding. There are two wheels or rotors and on each, only two rows of moving blades are attached because two row wheel are more efficient than three row wheel. In each wheel or rotor, velocity drops. There are two sets of nozzles in which whole pressure drop takes place i.e. whole pressure drop has been divided in small drops, hence it is pressure compounded.



Velocity compounded impulse turbine

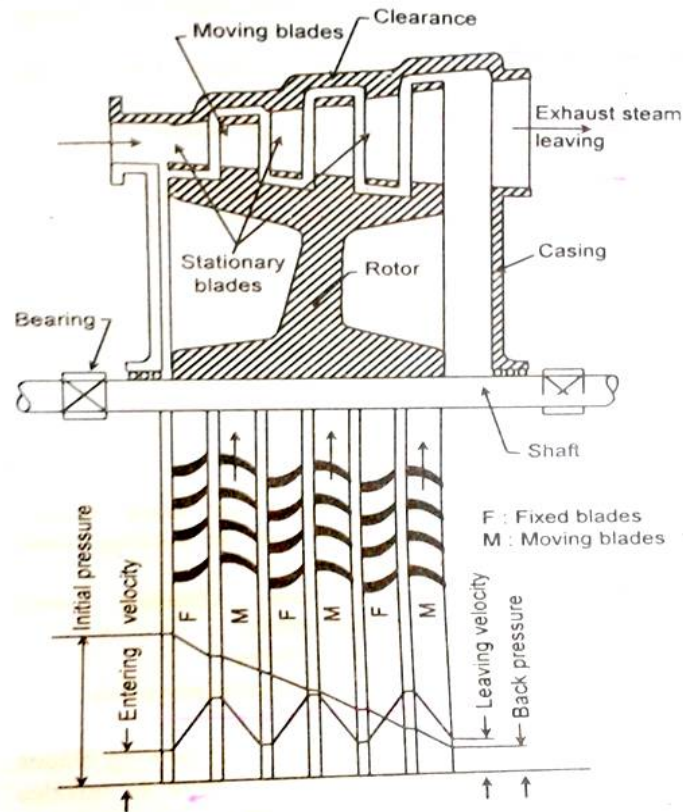
Pressure compounded impulse turbine



Pressure velocity compounded impulse turbine

2. Impulse –Reaction Turbines

This type of turbine utilizes the principle of impulse and reaction both. There are number of rows of moving blades attached to the rotor and an equal number of fixed blades attached to casing. The passage of moving blades is so designed that there is a small drop in pressure of steam in the moving blades which results in a increase in kinetic energy of steam. This kinetic energy gives rise to reaction in the direction opposite to that of added velocity. Thus, the gross propelling force or driving force is the vector sum of impulse and reaction forces.



Impulse- reaction turbine

QUESTIONS:

1. Classify steam turbines in details.
2. Why compounding is required in steam turbine?
3. Explain pressure compounded steam turbine with neat sketch.
4. Explain pressure velocity compounded steam turbine with neat sketch.
5. Explain reaction turbine with neat sketch.

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EXPERIMENT NO. 6

STUDY ABOUT DIFFERENT TYPES OF CONDENSERS AND COOLING TOWERS.

AIM:

To study about different types of condensers and cooling towers.

OBJECTIVE:

1. Study about the working of different types of condensers.
2. Study about the working of different types of condensers.

THEORY:

1. STEAM CONDENSERS.

A closed vessel in which steam is condensed by abstracting the heat from steam and the pressure is maintained below atmospheric pressure is known as condenser. The use of condenser in the power plant improves efficiency of the plant by decreasing the exhaust pressure of the steam below atmosphere. Steam condensed may be recovered to provide a source of good pure feed water.

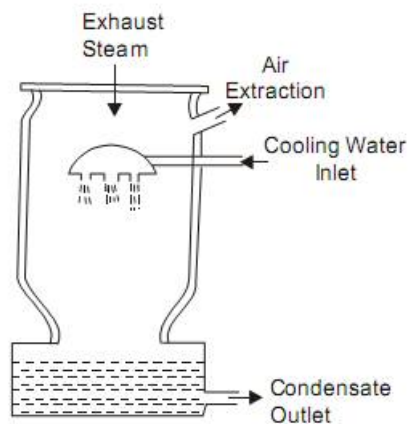
By using condenser efficiency of the plant increases as the enthalpy drop increases by increasing the vacuum in the condenser. Even the deposition of salt in the boiler is prevented with the use of condensate instead of using the feed water from outer source which contains salt.

TYPES OF STEAM CONDENSERS

The condensers are mainly classified as

A. Mixing type or Jet condensers.

In mixing type, the exhaust steam from prime mover and cooling water comes in direct contact with each other and steam condenses in water directly. The temperature of the condensate is same as that of cooling water leaving the condenser. They are preferred when good quality of water as feed to boiler is easily available in ample quantity.



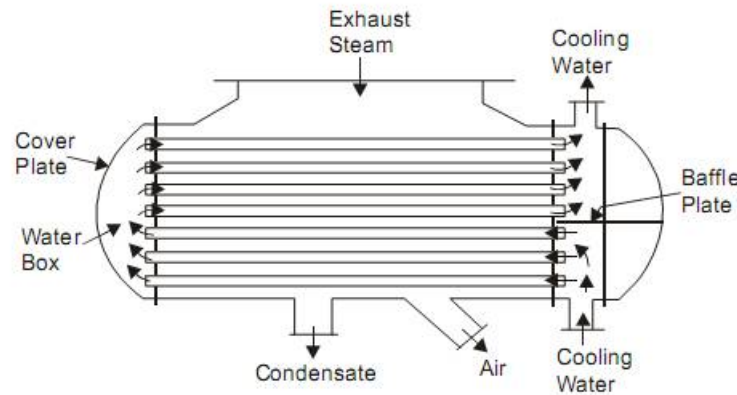
Jet Condenser

B. Non mixing type or Surface condensers.

In non mixing type, steam and cooling water do not come in direct contact with each other. The cooling water passes through the number of tubes attached to condenser shell and steam surrounds the tubes. These types of condensers are universally used in all high capacity modern steam power plants as the condensate coming out from the condenser is used as feed for the boiler.

The surface condensers are classified according to

- Number of water passes: single or multi pass
- Direction of condensate flow and tube arrangement: down flow and central flow.



Surface condenser

2. COOLING TOWERS.

The cooling water requirement in an open system is about 50 times the flow of steam to the condenser. A 1000 MW station will require about 100 thousand tons of circulating water per day even with the use of cooling towers.

As the cooling water takes the latent heat of steam in the condenser, the temperature of the water increases. The hot water coming out of the condenser cannot be used again in a closed system without pre cooling.

The cooling towers are divided mainly into two groups

1. Wet cooling towers:-

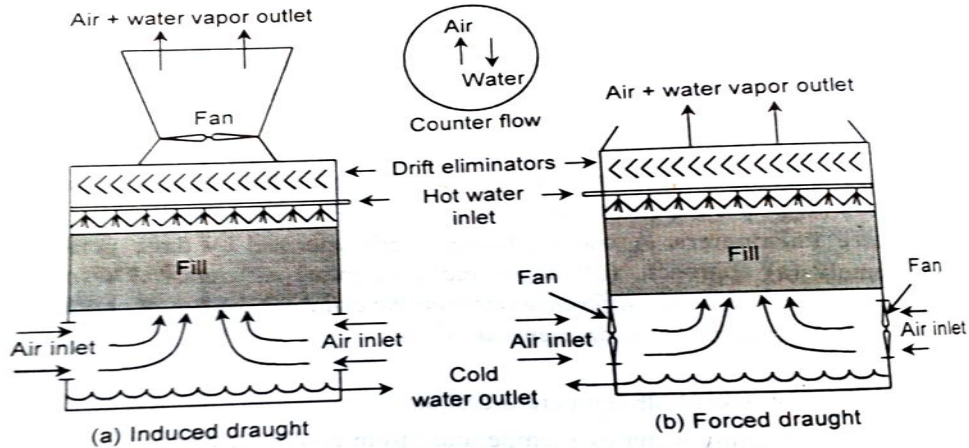
- Atmospheric or natural draught cooling towers

There is no fan used by natural draught cooling tower. They depend for air flow upon the natural driving pressure caused by the difference in density between the cool outside air and the hot humid air inside.

- Mechanical draught cooling towers.

The air is moved by one or more mechanically driven fans. The fan could be forced draught or induced draught type. The FD is mounted on the lower sides to force air into the tower while ID fan is located on the top of the tower.

The advantages of mechanical draught cooling towers include the assurance of moving the required quantity of air at all loads and climatic conditions. The main disadvantage includes power consumption, noise generation, operating and maintenance costs.



Mechanical draught cooling tower

2. Dry cooling towers:-

Dry cooling tower is one in which the circulating water is passed through finned tubes over which the cooling air is passed. All the heat rejected from the circulating water is in the form of sensible heat to the cooling air.

QUESTIONS:

1. Explain Evaporative condenser with its sketch.
2. Explain with neat sketch about surface condenser.
3. Explain jet condenser with neat sketch.
4. Explain deaeration in case of condenser.
5. Give the comparison between jet & surface condenser.
6. What is the requirement of cooling tower in steam power plant?

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EXPERIMENT NO. 7 TO STUDY ABOUT ECONOMICS OF POWER GENERATION SYSTEM.

AIM:

To study about economics of power generation systems.

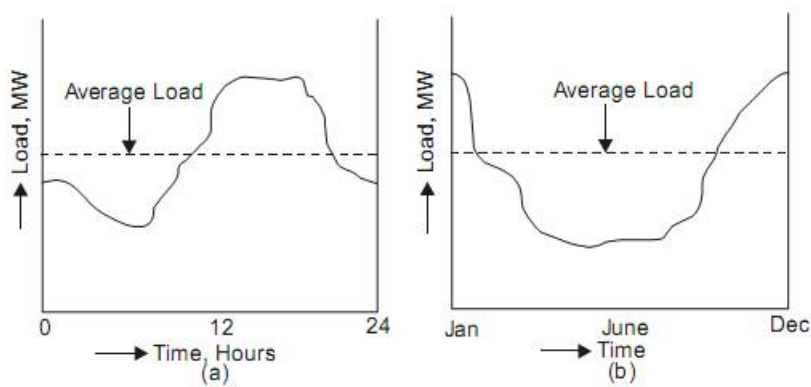
OBJECTIVE:

To study the cost analysis of power plant and different methods of tariff.

THEORY:

As we know that for operating any power plant we make an economic calculation to run effectively that power plant for the specific period of time & for that we need have to consider its main liabilities like its initial cost, operational cost, & other factors which helpful us to run the plant effectively within all aspects.

The load demand on a power system is governed by the consumers and for a system supplying industrial and domestic consumers, it varies within wide limits. This variation of load can be considered as daily, weekly, monthly or yearly. These curves are for a day and for a year and these show the load demanded by the consumers at any particular time. Such load curves are termed as “Chronological load Curves”. If the ordinates of the chronological load curves are arranged in the descending order of magnitude with the highest ordinates on left, a new type of load curve known as “load duration curve” is obtained.



a) Daily load curve and b) Yearly load curve

Cost of Power plant:

The cost analysis of power plant includes two types of costs.

1) Fixed cost:

The fixed cost is the capital invested in installation of complete plant. The cost includes the following

- a) Land cost
- b) Building cost
- c) Equipment cost
- d) Installation cost.
- e) Cost of transmission and distribution line.
- f) Depreciation cost
- h) Taxes and insurance premium

2) Running cost:

The running cost includes the following

- a) Fuel cost
- b) Labour cost
- c) Maintenance and repair cost
- d) Storage cost

Requirements of tariff:

Tariff should satisfy the following requirements:

- (1) It should be easier to understand.
- (2) It should provide low rates for high consumption.
- (3) It should encourage the consumers having high load factors.
- (4) It should take into account maximum demand charges and energy charges.
- (5) It should provide fewer charges for power connections than for lighting.
- (6) It should avoid the complication of separate wiring and metering connections.

Types of tariffs:

The various types of tariffs are as follows,

- (1) Flat demand rate
- (2) Straight line meter rate
- (3) Step meter rate
- (4) Block rate tariff
- (5) Two part tariff
- (6) Three part tariff.

QUESTIONS:

- 7. Explain Load curve and load duration curve.
- 8. List various types of costs involved in power plant and explain each in detail.
- 9. What is tariff? Explain various types of tariffs in detail.

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EXPERIMENT NO. 8

STUDY OF GAS POWER PLANT.

AIM:

To study of gas power plant.

OBJECTIVES:

4. Study about the basics of gas power plant.
5. Study about the brayton cycle and methods to improve efficiency power plant.

THEORY:

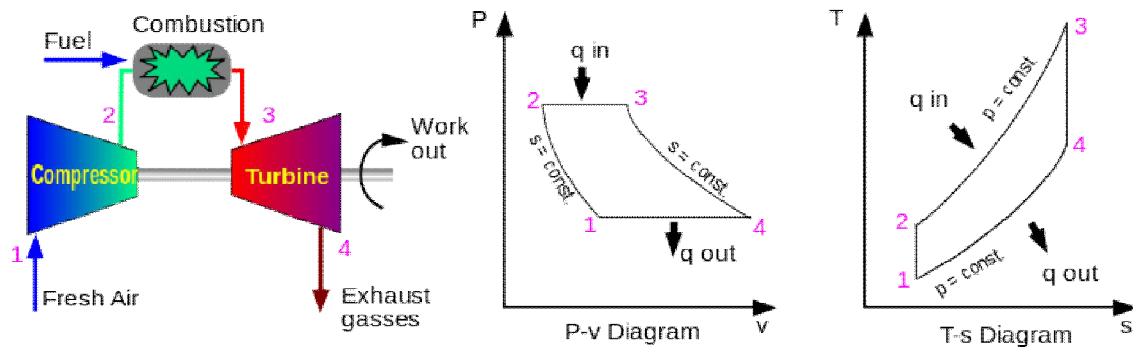
The gas turbine plant essentially consists of compressor, combustion chamber and turbine.

The air is compressed in a compressor and the fuel is burned in the combustion chamber when the compressed air is supplied from the compressor. The burned high temperature gases are passed through the turbine. The part of the work developed by the gases passing through the turbine is used to run the compressor and remaining (30-35%) is used to generate the electrical energy.

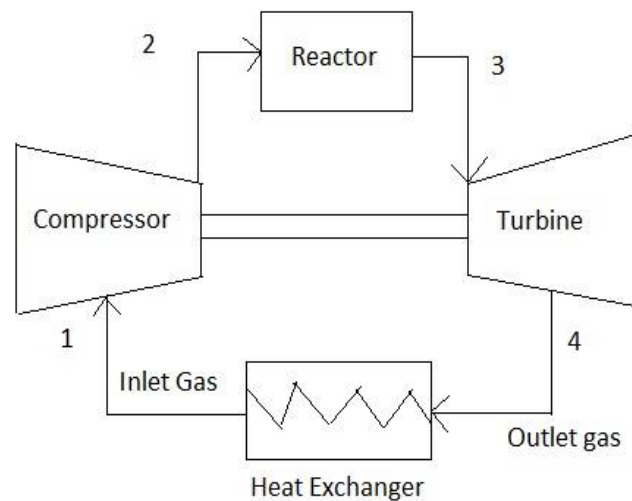
When the heat given to the air by mixing and burning the fuel in the air and the gases coming out of the turbine are exhausted to the atmosphere, the cycle is known as **open cycle power plant**. If the heat to the working medium (air or any other suitable gas) is given without directly burning the fuel in the air and the same working fluid is used again and again, the cycle is known as **closed cycle power plant**.

On the basis of combustion process the gas turbines are classified as

- (a) Continuous combustion or constant pressure type. The cycle working on this principle is called as Joule or Brayton cycle.
- (b) The explosion or constant volume type. The cycle working on this principle is known as Atkinson cycle.



Open cycle Gas turbine with Joule Brayton cycle.



Closed Cycle Gas turbine

Methods of improving efficiency and specific output of Gas Turbine Plant:-

1. Regeneration:- This is done by preheating the air with the turbine exhaust which saves the fuel consumption.
2. Improving turbine output by
 - Reheating- The whole expansion in turbine is achieved in two or more stages and reheating is done after each stage.
 - Increasing the value of maximum cycle temperature means turbine inlet temperature which requires
 - (a) Better quality of fuel
 - (b) New materials which can withstand high temperatures and pressures.
 - (c) Better blade cooling methods and cooling medium.
 - Improved turbine efficiency:- It depends on design improvements of each component.
 - Reducing compressor input. It may be done by
 - (a) Intercooling. Compressor work is reduced by intercooling the air between compressor stages.
 - (b) By lowering the inlet temperature to compressor. It is not possible because this will increase the pressure ratio.
 - (c) By increasing the compressor efficiency. This depends upon the design improvement.
 - (d) Water injection.

QUESTIONS:

1. Classify different types of gas turbines and their applications.
2. Explain working of simple open and closed gas turbine cycle.
3. Give the reasons that why actual Brayton cycle differs from the theoretical cycle.
4. Plot the various charts of following and explain the same
 - (a) Effect of regeneration, intercooling and reheating on efficiency.
 - (b) Effect of operating variable on thermal efficiency.
 - (c) Effect of operating variables on air rate.
 - (d) Effect of operating variables on work ratio.

(e) Water injection system.

REFERENCES:

1. Power Plant Engineering by P.K.Nag, TMH Publications.
2. Steam and Gas turbines and Power Plant Engineering by R.Yadav, Central publishing house, Allahabad.

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EXPERIMENT NO. 9

STUDY OF COMBINED STEAM & GAS TURBINE POWER PLANT.

AIM:

To study of combined steam & gas turbine power plant.

OBJECTIVES:

6. Study about the basics of combined cycle power plant.
7. Study about the parameters which affect the efficiency of combined cycle power plant

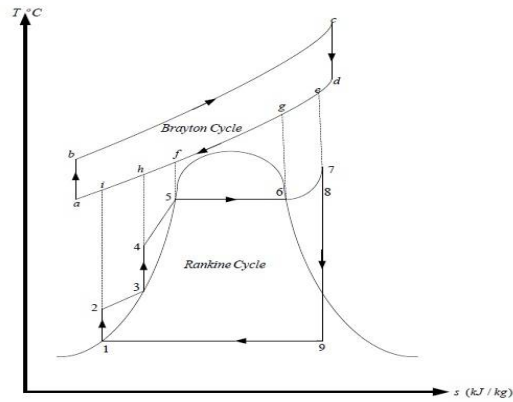
THEORY:

The gas turbine plants are having characteristics of quick starting and good response which make the gas turbine as desirable peak load and standby plant. The non availability of cooling water will not hamper gas turbine plant while running out steam turbine plant.

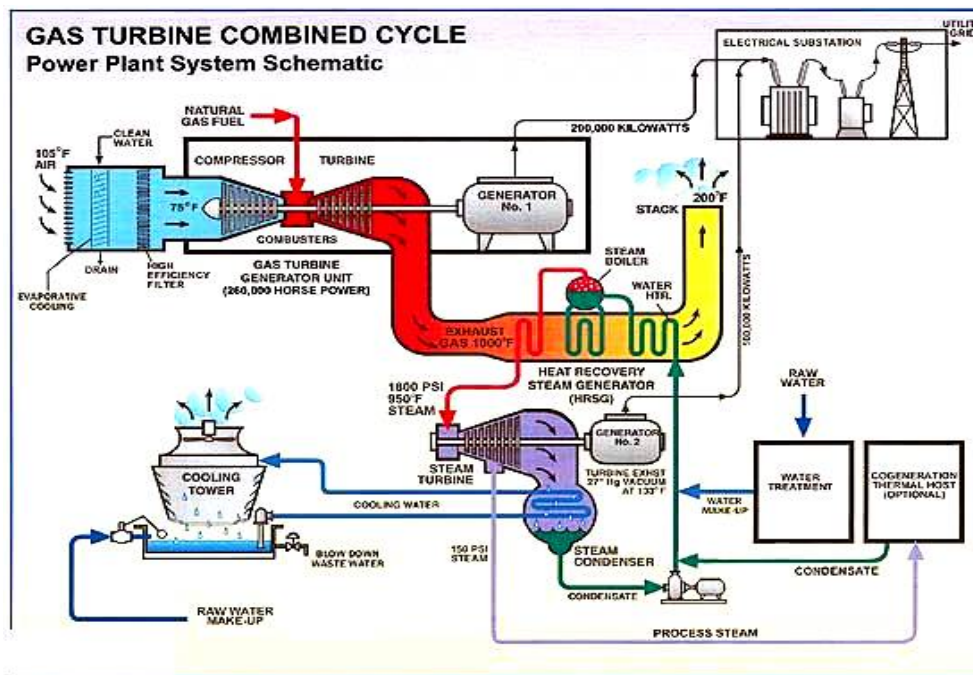
The temperature of the exhaust gases of a simple gas turbine plant lies between 400 to 500°C and contains about 16% oxygen compared with 21% in atmospheric air. A large quantity of energy (70% of initial) is also carried away by the exhaust gases with large quantity of oxygen without use.

An electrical utility industry has launched an effort to recover the heat energy of the exhaust gases by coupling a steam plant with a gas turbine installation. This combined cycle recovers much of exhaust energy by passing high temperature exhaust gases to heat recovery boiler to generate steam which can be further used to drive a steam turbine.

Increased power and high thermal efficiency obtained from this, the concept of combined cycle reduces the cost of the additional equipment and lowers the generating cost if the number of operating hours per year substantially increased.



T-S representation of Combined cycle power plant



Combined steam and gas cycle power plant

Heat Recovery Steam Generator

Heat Recovery Steam Generator (HRSG) is a boiler that utilizes heat energy of residual exhaust gases from gas turbine unit to heat water and convert it into steam, and steam is then used to drive steam turbines.

Heat Recovery Steam Generator (HRSG) boiler is not equipped with burners and do not consume fuel, so there is not radiation heat transfer process.

The heat transfer process that occurs only convection and conduction from exhaust gas of gas turbine into water to be processed into steam through heating elements inside HRSG boiler room.

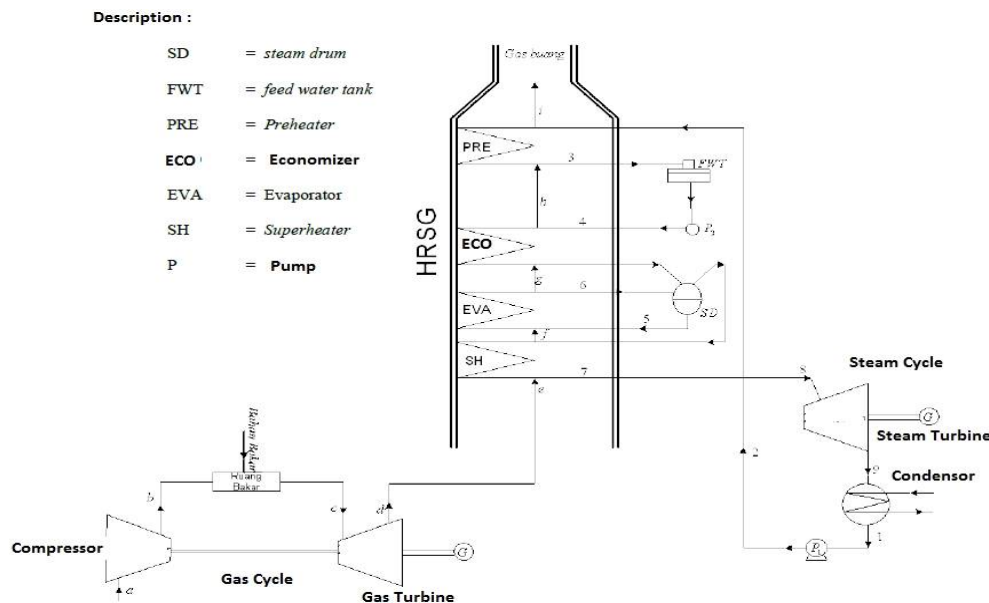


Diagram of CCPP and HRSG (Heat Recovery Steam Generator) Single Pressure

QUESTIONS:

1. Explain the working of combined cycle.
2. Explain the different parameters which affect the efficiency of combined cycle.
3. Explain heat recovery steam generator (**HRST**)

REFERENCES:

1. Steam and Gas turbines and Power Plant Engineering by R.Yadav, Central publishing house, Allahabad
2. Power Plant Engineering by P.K.Nag, TMH Publications.

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EXPERIMENT NO. 10

TESTING OF DIESEL FIRED WATER TUBE BOILER BASED STEAM POWER PLANT.

AIM:

Testing of diesel fired water tube boiler based steam power plant.

OBJECTIVE:

To study the performance of steam power plant and finding different parameters of different components.

THEORY:**OBSERVATION TABLES AND CALCULATIONS:****1. Calorimeter**

Let P1 and T1 = Pressure and temperature in separating chamber

P2 and T2 = Pressure and temperature in throttling chamber

Mt = Steam throttled to throttling calorimeter

Ms = Moisture separated in separating calorimeter.

X1 = Steam quality in separating calorimeter

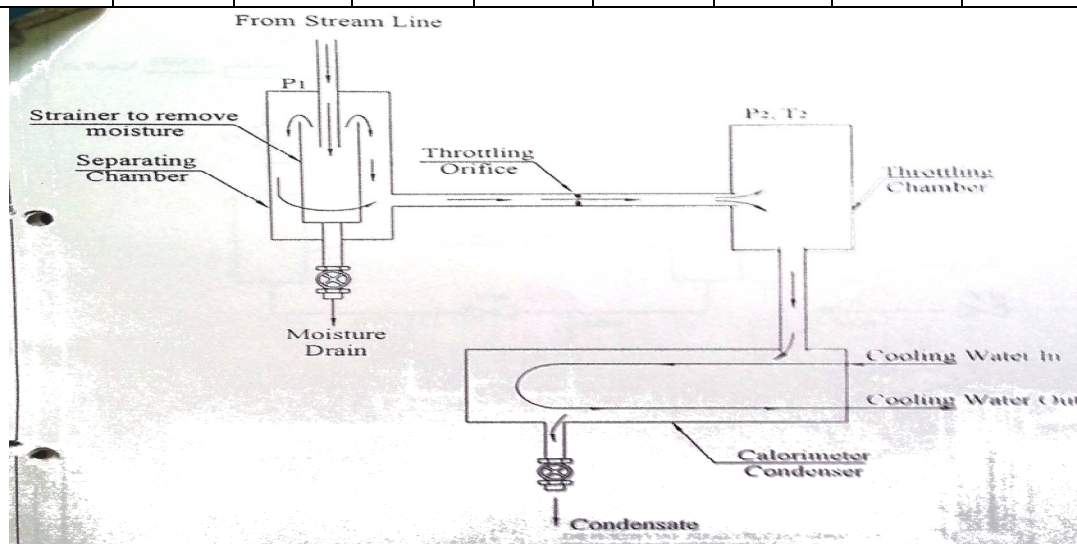
Mw = Total moisture in steam sample

Mc = Quantity of water collected in separating calorimeter.

t = time required to measure Mc

Sr..No.	P1	T1	t	Mc	P2	T2	Hf1	Hfg1	H2
	Kg/sq cm	°c	minute	Cc	Kg/sq cm	°c	Kcal/kg	Kcal/kg	Kcal/kg

1									
2									



$$M_t = 2.5 * P_1$$

$$M_s = \frac{(M_c * 60)}{t * 1000}$$

$$M_w = (M_s + (1 - X_1)M_t)$$

$$X_1 = \frac{(h_2 - h_{f1})}{h_{fg1}}$$

$$\text{Steam Quality } X = \left(1 - \frac{M_w}{(M_s + M_t)}\right) * 100\%$$

Sr..No.	X1	Ms Kg/hr	Mt Kg/hr	Mw Kg/hr	X
1					
2					

2. Flow meter

Here we will use the condensate directly from the condenser drain as steam flow rate M.

3. Isentropic work calculations

P_E and T_E be the pressure and temperature at exit

Sr.No.	P0 Kg/sq cm	T0 °C	X	h0 Kcal/kg	S0 Kcal/kg°C	PE Kg/sq cm	XE	hE Kcal/kg	W kw
1									
2									

$$h_0 = h_{fo} + Xh_{fgo}$$

$$S_0 = S_{fo} + XS_{fgo}$$

$$X_E = \frac{(S_0 - S_{fe})}{S_{fgE}}$$

$$h_s = h_0 - (h_{fe} + X_E h_{fgE})$$

$$\text{Isentropic work } W = \frac{M * h_s}{860.6} \text{ KW}$$

4. Output power calculations

Let V = voltmeter reading in volts

A =ammeter reading in amps

$$\text{Alternator output power} = \frac{V * A}{1000}$$

$$\text{Steam turbine output} = \frac{\text{Alternator output}}{0.65}$$

Where 0.65 is alternator efficiency

Sr.No.	Generator Output V volts	Generator Output A amp	Generator Output Kw	Turbine speed N rpm	Turbine output power kw

1					
2					

Calculation of isentropic efficiency of turbine:-

$$\eta_{isen} = \frac{\text{Turbine Power}}{\text{Isentropic work}}$$

5. Condenser calculations

Coolant water flow rate-

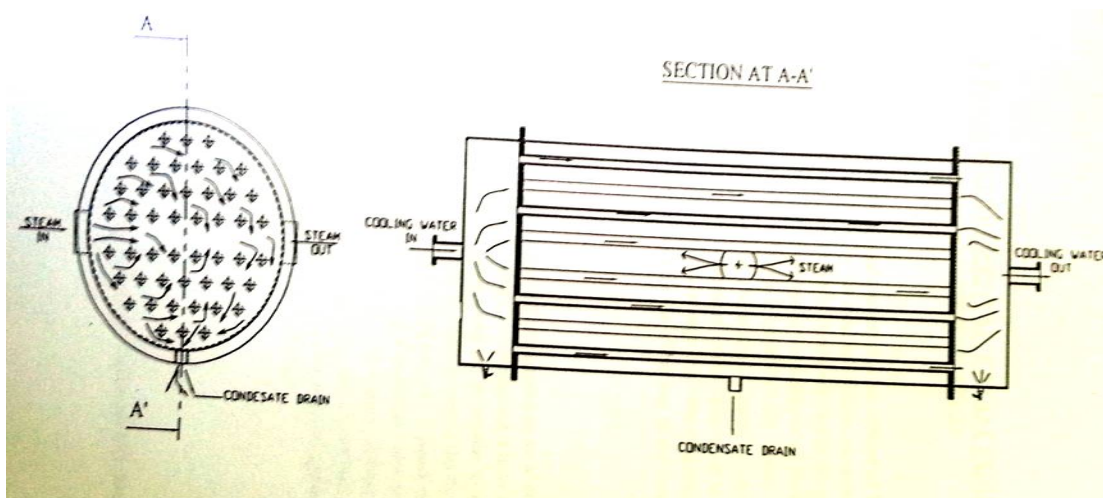
The orifice as been calibrated

Orifice pressure gauge reading 1 =OP1

Orifice pressure gauge reading 2 =OP2

Difference in pressure = (OP1-OP2) kg/sq.cm

OP = 10(OP1-OP2) m of water



Sr.No	OP1	OP2	Mc	TC1	TC2	H	TE	Effectiveness
	Kg/sq cm	Kg/sq cm	Kg/min	°c	°c	Kcal/kg	°c	

1								
2								

Coolant water flow rate-

The orifice as been calibrated

Orifice pressure gauge reading 1 =OP1

Orifice pressure gauge reading 2 =OP2

Difference in pressure = (OP1-OP2) kg/sq.cm

OP = 10(OP1-OP2) m of water

Coolant water flow rate Q = 0.00204 $\sqrt{OP}$ m3/sec

Coolant mass flow rate Mc =60*1000*Q kg/min

Coolant water inlet temperature = TC1 °C

Coolant water exit temperature = TC2 °C

Heat removed by coolant water $H = M_c * (T_{c1} - T_{c2})$

Heat exchanger effectiveness of condenser $\epsilon = \frac{(T_{c2} - T_{c1})}{(T_E - T_{c1})}$

EFFICIENCY OF BOILER USING DIRECT METHOD:

Heat input data

Quantity of diesel consumed (q) :

GCV(Gross calorific value) of Diesel :

Heat output data

Quantity of steam generated (Q) :

Steam pressure(P)/temperature :
Enthalpy of steam at P (H) :
Feed water temperature :
Enthalpy of feed water(h) :

$$\text{Boiler efficiency}(\eta) = \frac{Q^*(H - h)}{q^* GCV} * 100$$

CONCLUSION:

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